Determining the Condition Number of a Real or Complex Matrix from Its Submatrices

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Abstract

The condition number of a matrix governs how perturbations in the input affect the accuracy of computed solutions, determining how many digits are lost in finite-precision arithmetic. While many equivalent definitions exist, explicit structural decompositions are rare. This paper presents a new block decomposition identity for the Frobenius-norm condition number $\kappa_F(A_{n+1})$. The formula expresses κ_F exactly as the product of two four-term factors, yielding a 4×4 grid of contributions that localize which subblocks (A_n, V_1, V_2, b_0) dominate ill-conditioning. The theorem is established through a complete mathematical proof and validated numerically: results match the direct Frobenius condition number to machine precision across 100 random 100×100 test matrices. Numerical examples, including near-singular cases, further confirm the analysis. Short remarks connect the result to applications in high-performance computing (HPC), machine learning (ML), artificial intelligence (AI), and mathematical finance (q-fin.MF). In particular, the decomposition clarifies conditioning patterns in large covariance matrices used for portfolio optimization and risk modeling. To our knowledge, this is the first exact decomposition of the Frobenius condition number in block form.

1 Introduction

The condition number of a matrix is a classical measure of sensitivity in numerical linear algebra. It determines how perturbations in the input propagate to perturbations in the solution and how many digits of accuracy remain in finite precision arithmetic.

Most computers today, including high–performance computing systems, are still based on the 1985 IEEE 754 floating–point standard, which provides about sixteen decimal digits of precision. In the decades since, the demands placed on this standard have grown sharply, while advances in managing matrices with large condition numbers have lagged. As a result, many of today's most important computational tools are running at the edge of numerical stability.

The growth of the World Wide Web, HPC, machine learning, and artificial intelligence has brought matrices to the center of modern computation. Examples range from classical algorithms such as Linear programming, Floyd–Warshall for shortest paths, Gradient Descent, Google rank

algorithm, Hessian matrices in optimization, graph–theoretic adjacency structures, to the ubiquitous softmax function in AI models. In science and engineering, matrices are now the fundamental units of data storage and processing.

In engineering practice, stress–strain relations for orthotropic composite materials are governed by matrices, and here an exact formula for the condition number is directly relevant. In computational fluid dynamics, finite element methods, CAD/CAM, and simulation, the fidelity of results hinges on matrix calculations that remain stable at scale. Across these fields, improved understanding and management of conditioning is essential.

While many equivalent definitions of norms and condition numbers exist, the Frobenius–norm condition number offers a useful balance between theoretical clarity and computational accessibility, and it is the focus of this work.

The main contribution of this paper is a new block decomposition identity for $\kappa_F(A_{n+1})$, stated in Theorem 1. This identity localizes the growth of the condition number into a 4×4 structure, showing which subblocks (A_n, V_1, V_2, b_0) dominate ill conditioning. Numerical examples confirm the analysis, and short remarks highlight relevance to modern high–performance computing, optimization, and machine learning. To our knowledge, this is the first exact decomposition of the Frobenius condition number in block form. Some areas of research resulting from Theorem 1 with the main definitions below:
Beyond the scope of this note, Theorem 1 suggests several natural directions for future research and development:

1. **How do the sixteen cross–terms** from the product of the two four–term factors interact in practice, and which dominate in different matrix regimes?

2. What structural properties of the block L_n (depending on m_0, the outer product V_1V_2, and A_n^{-1}) control its contribution to $\kappa_F(A_{n+1})$?

3. How do the edge vectors V_1 and V_2, and their products V_2V_1 and V_1V_2, influence sensitivity and stability?

4. **What are the roles of the scalars k_0, m_0, and b_0**, especially when m_0 becomes unbounded as b_0 approaches k_0 (cf. the summary of definitions below)?

5. How does the decomposition specialize for structured classes of matrices (symmetric, triangular, sparse, orthogonal, Hermitian)?

6. Which applied matrices arising in engineering, computer science, and related fields such as machine vision, AI, large language models, and robotics exhibit distinctive blockwise conditioning patterns?

7. **What inequalities of norms can be applied to the sixteen terms** to derive practical bounds on κ_F? How are other matrix norms (spectral, 1-, or ∞-norms) related to this result?

8. How can numerical exploration and software development based on this decomposition support applications in engineering/science, leading to faster algorithms and deeper understanding of sensitivity and stability? Porting code to CS, Python, C++ with more capable data types.

Together, these directions underline that Theorem 1 is not only an exact identity but also a starting point for broader exploration of conditioning in theory, computation, and applications.

$$k_0 := V_{2n}A_n^{-1}V_{1n} \qquad m_0 := \frac{1}{b_0 - k_0},$$

$$V := V_{1n}V_{2n} \in \mathbb{K}^{n\times n} \text{ (rank-1)} \qquad L_n := I_n + m_0\, V\, A_n^{-1}$$

The Canonical (Natural) Partition of a Square Matrix, Named CANMXP, into a 2×2 Block Matrix

We will see the CANMXP procedure below to express any square matrix of n+1 elements into a natural 2x2 block matrix many times in this research; therefore, here we cover it in detail.

Definition 1 (The Natural Matrix Partition function N_{MP})**.** *We define*

$$N_{\text{MP}} :\ \mathcal{M}(n{+}1) \ \longrightarrow\ \mathcal{MP}(2),$$

where $\mathcal{M}(n{+}1)$ is the set of real/complex square matrices of size $(n{+}1) \times (n{+}1)$ and $\mathcal{MP}(2)$ is the set of real/complex 2×2 block *matrices. The map N_{MP} sends any $(n{+}1) \times (n{+}1)$ matrix to a natural (canonical) 2×2 block form.*

Let $A_{n+1} = (b_{ij})$ be an $(n{+}1) \times (n{+}1)$ matrix with $i, j = 1, \ldots, n{+}1$.

Definition 2 (Top-left $n \times n$ principal block)**.** *Define $A_n = (a_{ij})$ as the $n \times n$ matrix with $i, j = 1, \ldots, n$ such that $a_{ij} = b_{ij}$ for all $i, j \in \{1, \ldots, n\}$; equivalently, we may write $A_n = (b_{ij})_{i,j=1}^{n}$. We write $A_n \subseteq A_{n+1}$ to indicate that A_n is* contained *in A_{n+1} as its top-left principal submatrix.*

Definition 3 (Right "edge" vector)**.** *The vector V_{1n} (also written V_1) is the right column adjoining A_n:*

$$V_{1n} = V_1 = (\, b_{i,\,n+1}\,)_{i=1}^{n} \in \mathbb{K}^{n\times 1}.$$

Definition 4 (Bottom "edge" vector)**.** *The vector V_{2n} (also written V_2) is the bottom row adjoining A_n:*

$$V_{2n} = V_2 = (\, b_{n+1,\,j}\,)_{j=1}^{n} \in \mathbb{K}^{1\times n}.$$

Definition 5 (Last diagonal element)**.** *The lower right pivot is*

$$D_n = (\, b_{n+1,\,n+1}\,) = b_{n+1,\,n+1} \in \mathbb{K},$$

a matrix / scalar 1×1. Note that D_n is not part of any edge vector V_{1n} or V_{2n}, nor of the block A_n.

Note that we also often refer to Theorem 1 below (by a latex counter) as Theorem 6.2 - but in the original research paper in 10/29/2010 by Raul the notes were in a (unpublished) PowerPoint paper and the steps to the proofs were spelled before what is now Theorem 1, thus the 6.2.

In summary, the basic idea is simple:
We create 4 blocks, the main submatrix is a nxn block named: A_n, the bottom edge becomes a row vector - a 1xn block named V1 or V_{1n}, the right side becomes a column vector - a nx1 block named V2 or V_{2n}, and finally the last diagonal element as a single block of size 1x1, named b0 or b_0. The sum of elements is then n x n + n + n + 1 = (n+1) x (n+1).

So we define this mapping through a formal function $N_{\text{MP}}()$. Here we basically use the enormous simplification of converting a real/complex square matrix of arbitrary size (say n+1 x n+1) to a 2×2 block matrix.

Then in the following, we use this natural decomposition (CANMXP) of a square matrix into a simpler 4x4 block matrix, which enables us to prove the main Theorem 1 on condition numbers.

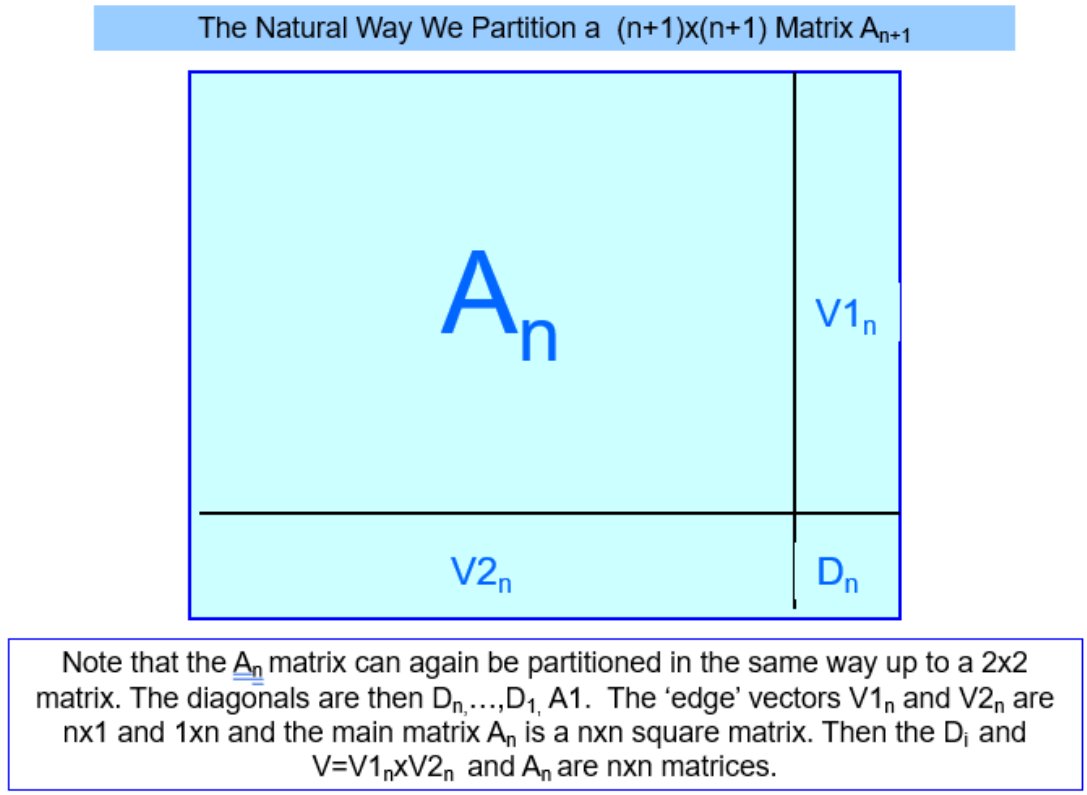


Figure 1: Natural partition of an $(n+1)\times(n+1)$ matrix A_{n+1} into a 2×2 block form. The main block is A_n, the "edge" vectors are V_1 and V_2, and the lower-right scalar is the pivot b_0 (also written D_n in the diagram). The block A_n can itself be partitioned recursively.

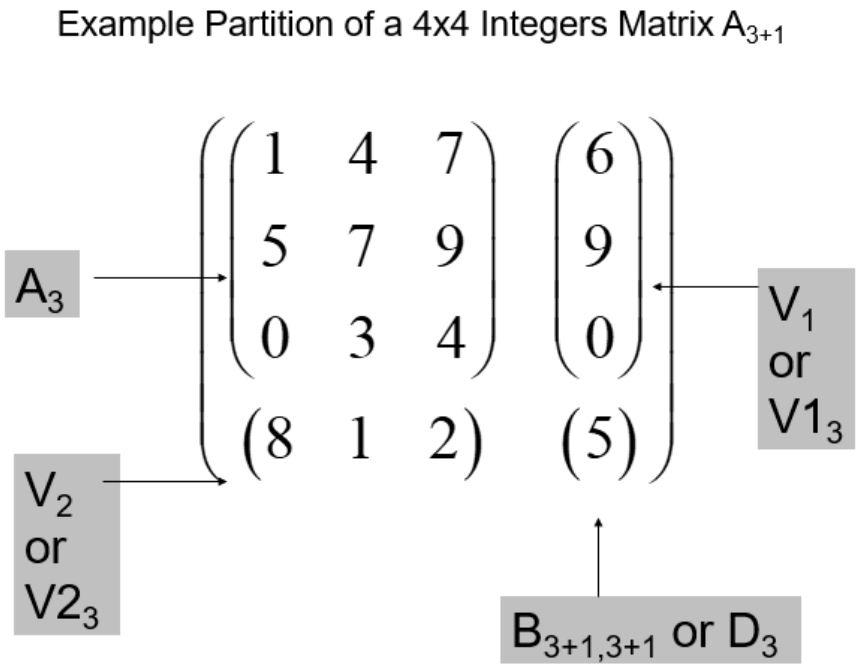
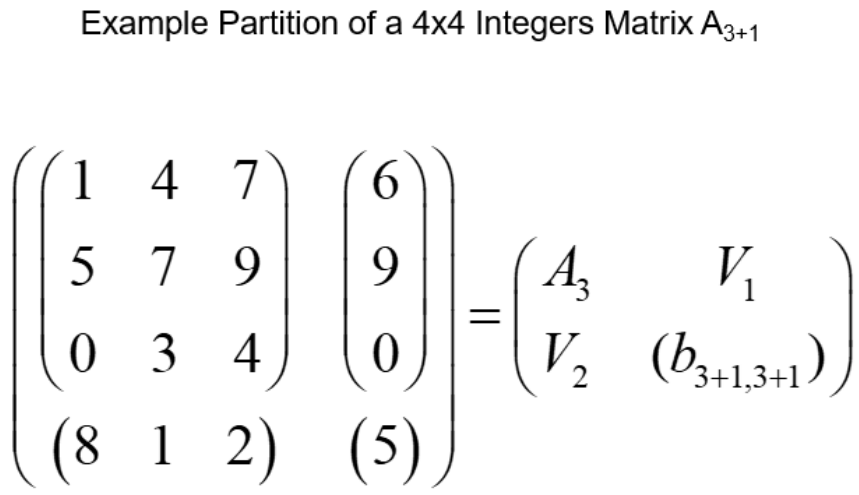


Figure 2: The same example with block labels A_3, V_1, V_2, b_0.



$$\begin{pmatrix} \begin{pmatrix} 1 & 4 & 7 \\ 5 & 7 & 9 \\ 0 & 3 & 4 \end{pmatrix} & \begin{pmatrix} 6 \\ 9 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 8 & 1 & 2 \end{pmatrix} & \begin{pmatrix} 5 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} A_3 & V_1 \\ V_2 & (b_{3+1,3+1}) \end{pmatrix}$$

Figure 3: Equality view: $A_{3+1} = (A_3, V_1; V_2, b_0)$.

We state the main results without proof to make clear what we are aiming for:

Field and Basic Notation

$\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ (we keep the base field abstract). For matrices/vectors we use the Frobenius/Euclidean norm; for scalars $|\cdot|$ denotes absolute value (modulus over $\mathbb{C}$).

Definition of Frobenius Norm and CONDITION Number for a NxN Real or Complex Matrix

$$\|A\|_F = \Big(\sum_{i=1}^{n}\sum_{j=1}^{n} |a_{ij}|^2\Big)^{1/2} = \sqrt{\mathrm{tr}(A^*A)} \qquad (A \in \mathbb{C}^{n\times n}).$$

$$\kappa_F(A) = \begin{cases} \|A\|_F\, \|A^{-1}\|_F, & A \in \mathbb{C}^{n\times n} \text{ invertible}, \\ \infty, & \text{otherwise}. \end{cases}$$

$$\kappa_F^{\mathrm{LS}}(A) := \|A\|_F\, \|A^{+}\|_F \qquad (A \in \mathbb{C}^{n\times n}).$$

$$\mathbf{cond}(A) := \|A\|_{\mathbf{F}} \|A^{-1}\|_{\mathbf{F}}.$$

Using the definition above, the Frobenius condition number of A_{n+1} is:

Notation for the Main Theorem 1 Named: FCNFACT

Let $A_{n+1} = \left[\begin{smallmatrix} A_n & V_{1n} \\ V_{2n} & b_0 \end{smallmatrix}\right]$ with $A_n \in \mathbb{K}^{n\times n}$ invertible, $V_{1n} \in \mathbb{K}^{n\times 1}$ $V_{2n} \in \mathbb{K}^{1\times n}$ $b_0 \in \mathbb{K}$. Define

$$k_0 := V_{2n} A_n^{-1} V_{1n} \qquad m_0 := \frac{1}{b_0 - k_0},$$

$$V := V_{1n} V_{2n} \in \mathbb{K}^{n\times n} \text{ (rank-1)} \qquad L_n := I_n + m_0\, V\, A_n^{-1}$$

We write $\|\cdot\|_F$ for the Frobenius norm and $\mathrm{cond}(A) := \|A\|_F \|A^{-1}\|_F$.

Theorem 1 (Main identity for the Frobenius condition number)**.** *Let*

$$A_{n+1} = \begin{pmatrix} A_n & V_1 \\ V_2 & b_0 \end{pmatrix},$$

with $A_n \in K^{n\times n}$ invertible, $V_1 \in K^{n\times 1}$, $V_2 \in K^{1\times n}$, $b_0 \in K$. Define

$$k_0 := V_2 A_n^{-1} V_1, \quad m_0 := \frac{1}{b_0 - k_0}, \quad L_n := I_n + m_0 (V_1 V_2) A_n^{-1}.$$

Then

$$\kappa_F(A_{n+1}) = \sqrt{\|A_n\|_F^2 + \|V_1\|_2^2 + \|V_2\|_2^2 + |b_0|^2}\, \sqrt{\|A_n^{-1} L_n\|_F^2 + |m_0|^2\, (\|A_n^{-1} V_1\|_2^2 + \|V_2 A_n^{-1}\|_2^2 + 1)}.$$

Notes on the Variables Above:

The variables above used inside F and G are all based on the Canonical (Natural) Partition of a Square Matrix - into a 2 × 2 Block Matrix, that we described before.
In Summary the V1, V2 vectors are the EDGE vectors of the sub-matrix An is the sub-matrix of An+1, b0 is the last diagonal element (a 1x1 matrix), Ln is defined as the identity matrix + m0 x V x the inverse of An, where V = V1 * V2 a nxn matrix, and m0 = 1/(b0 - k0) a scalar, and finally k0 is defined as V2 * inverse of An * V1. Note V1 * V2 is nxn but V2 * V1 is 1x1 a scalar.

Why do we use the Frobenius CONDITION Number?

Remark 1 (Finite precision and loss of digits). In IEEE 754 double precision (binary64), the unit roundoff is $\varepsilon \approx 1.11 \times 10^{-16}$, corresponding to about 16 decimal digits. For backward–stable algorithms the forward error scales as $\mathcal{O}(\kappa(A)\,\varepsilon)$, so the number of trustworthy digits is approximately

$$16 \;-\; \log_{10}\big(\kappa(A)\big).$$

Thus if $\kappa(A) \approx 10^8$, only about 8 digits of accuracy remain. This simple rule of thumb motivates the need to analyze κ and, in particular, to localize its growth via Theorem 1.

Theorem 2 (Norm–equivalence of condition numbers)**.** *Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be two consistent matrix norms on $\mathbb{K}^{n\times n}$, and let*

$$\kappa_a(A) := \|A\|_a\,\|A^{-1}\|_a, \qquad \kappa_b(A) := \|A\|_b\,\|A^{-1}\|_b$$

for any invertible $A \in \mathbb{K}^{n\times n}$. Then there exist constants $c, C > 0$, depending only on n and the choice of norms, such that

$$c\,\kappa_a(A) \;\le\; \kappa_b(A) \;\le\; C\,\kappa_a(A) \qquad \textit{for all invertible } A.$$

Remark 2. This theorem is a direct consequence of norm equivalence in finite dimensions: all vector norms on $\mathbb{K}^n$ are equivalent, and so are their induced matrix norms. In particular, for $n \times n$ matrices one has explicit constants, for example

$$\kappa_2(A) \;\le\; \kappa_F(A) \;\le\; n\,\kappa_2(A), \qquad \kappa_2(A) \;\le\; \kappa_1(A),\, \kappa_\infty(A) \;\le\; n\,\kappa_2(A),$$

which justify the use of the Frobenius condition number as an equivalent measure (up to constants) to the spectral or operator condition numbers.

A major reason also is that the definition of this condition number, based on the Frobenius NORM is easily computable directly from the matrix entries as the square root of the sum of all the absolute values of each matrix element, for both Real/Complex square matrices, and more important, this norm allows all the needed manipulations needed to prove all the preliminary results to the main Theorem 1 - FCNFACT above.
Also the Frobenius norm for the matrix inverse is the same definition, but for the entries of the inverse matrix. In Matlab code (M-files) this condition number is defined as: cond(A,'fro') for matrix A, where 'fro' indicates the Frobenius norm, which is also denoted as norm(A, 'fro').

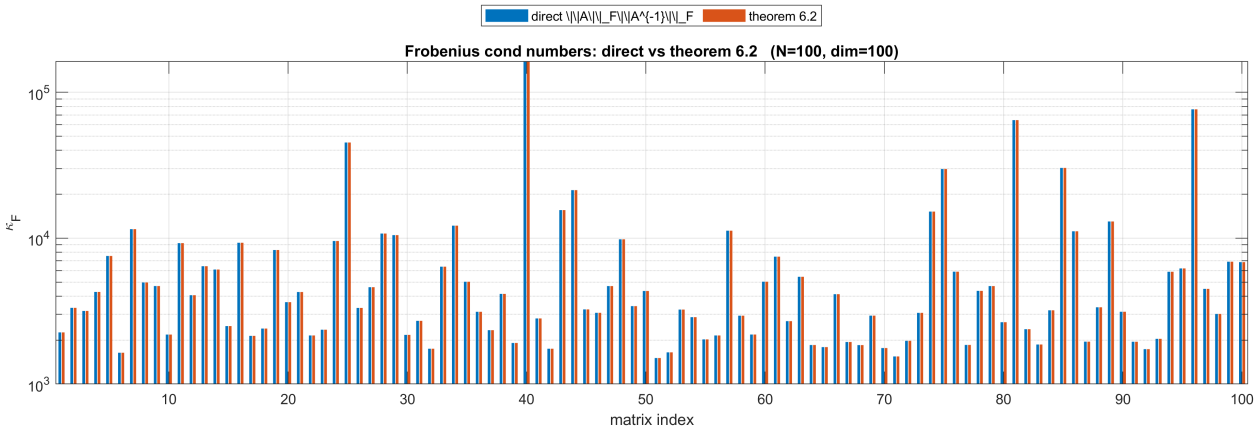


Figure 4: Matlab Plot Bar of conf(M,'fro') vs Theorem 1 - Shows all 100 bars are equal

Informal Summary of Theorem 1 - Named FCNFACT

Divide A_{n+1} into the 2x2 block matrix $\{A_n, V_{1n}, V_{2n}, b_0\}$. Measure the sizes (Frobenius norm) of these four building blocks and measure the matching inverse side sizes driven by A_n^{-1} and m_0.
The condition number equals the product of these two four-term square roots. This decomposition lets us pinpoint which block(s) dominate ill-conditioning, out of 4x4 or 16 products.

Sixteen-Term Breakdown (interpretation of the product)

The product of the two four-term square roots expands into 4×4 interaction terms C_{ij}. Rows correspond to $\{A_n, V_{1n}, V_{2n}, |b_0|\}$; columns to $\{A_n^{-1}L_n,\ m_0A_n^{-1}V_{1n},\ m_0V_{2n}A_n^{-1},\ m_0\}$. Each C_{ij} is the product of a "direct" squared norm with an "inverse-side" squared norm.

Table 1: Compact 4×4 table of C_{ij} (products of squared norms). The sum equals $\text{cond}(A_{n+1})^2$.

Row \ Column	$\mathbf{A_n^{-1}L_n}$	$\mathbf{m_0A_n^{-1}V_{1n}}$	$\mathbf{m_0V_{2n}A_n^{-1}}$	$\mathbf{m_0}$
$\mathbf{A_n}$	$\|A_n\|_F^2\|A_n^{-1}L_n\|_F^2$	$\|A_n\|_F^2\|m_0A_n^{-1}V_{1n}\|_F^2$	$\|A_n\|_F^2\|m_0V_{2n}A_n^{-1}\|_F^2$	$\|A_n\|_F^2\|m_0\|^2$
$\mathbf{V_{1n}}$	$\|V_{1n}\|_F^2\|A_n^{-1}L_n\|_F^2$	$\|V_{1n}\|_F^2\|m_0A_n^{-1}V_{1n}\|_F^2$	$\|V_{1n}\|_F^2\|m_0V_{2n}A_n^{-1}\|_F^2$	$\|V_{1n}\|_F^2\|m_0\|^2$
$\mathbf{V_{2n}}$	$\|V_{2n}\|_F^2\|A_n^{-1}L_n\|_F^2$	$\|V_{2n}\|_F^2\|m_0A_n^{-1}V_{1n}\|_F^2$	$\|V_{2n}\|_F^2\|m_0V_{2n}A_n^{-1}\|_F^2$	$\|V_{2n}\|_F^2\|m_0\|^2$
$\mathbf{\|b_0\|}$	$\|b_0\|^2\|A_n^{-1}L_n\|_F^2$	$\|b_0\|^2\|m_0A_n^{-1}V_{1n}\|_F^2$	$\|b_0\|^2\|m_0V_{2n}A_n^{-1}\|_F^2$	$\|b_0\|^2\|m_0\|^2$

Remark 3 (Empirical cross-check - Theorem 1 Numerical Verification). For 100 test random 100×100 matrices, each matrix of the 100 matrices has 10,000 numbers for a total of 1,000,000 numbers - the closed form in Theorem 1 matches the direct Frobenius condition number, with machine precision (median relative error $\approx 10^{-13}$); Using MATLAB Ver R2024b 16 digits of precision and Mersenne Twister for random number generation.
Before we start the proof of Theorem 1 please recall the definition of the Frobenius Norm and Condition number as:

$$\|A\|_F = \Big(\sum_{i=1}^{n}\sum_{j=1}^{n}|a_{ij}|^2\Big)^{1/2}$$

	A	B	C	D	E
1	Log of Tests of Cond Number Theorem 1-or 6.2 (col C) vs actual Frob Norm (col B) - to 5 dec (col D) to 12 dec (col E with Data Bar)				
2	run	kappa_direct-12 dec places	kappa_theo62-12 dec places	DIFFERENCES-7 dec places	DIFFERENCES-12 dec/data bars
3	1	2,260.186291449740	2,260.186291449740	0.0000000	0.000000000000
4	2	3,329.136845858260	3,329.136845858200	0.0000000	0.000000000060
5	3	3,170.536842870330	3,170.536842870350	0.0000000	-0.000000000020
6	4	4,279.079922907620	4,279.079922907630	0.0000000	-0.000000000010
7	5	7,546.159151221510	7,546.159151221390	0.0000000	0.000000000120
8	6	1,638.993923677630	1,638.993923677630	0.0000000	0.000000000000
9	7	11,528.986453916200	11,528.986453916200	0.0000000	0.000000000000
10	8	4,966.446323497000	4,966.446323496930	0.0000000	0.000000000070

Theorem 1 Numerical Verification - proof below

$$\kappa_F(A) = \begin{cases} \|A\|_F \, \|A^{-1}\|_F, & A \in \mathbb{C}^{n\times n} \text{ invertible}, \\ \infty, & \text{otherwise.} \end{cases}$$

Theorem 3 ((Frobenius norm of A_{n+1} in NMP form)). *Let*

$$A_{n+1} \;=\; \begin{pmatrix} A_n & V1_n \\ V2_n & b_0 \end{pmatrix}, \qquad A_n \in \mathbb{K}^{n\times n},\; V1_n \in \mathbb{K}^{n\times 1},\; V2_n \in \mathbb{K}^{1\times n},\; b_0 \in \mathbb{K}.$$

Then

$$\boxed{\|A_{n+1}\|_F = \sqrt{\|A_n\|_F^2 + \|V1_n\|_2^2 + \|V2_n\|_2^2 + |b_0|^2}\;.} \tag{1}$$

Proof. View A_{n+1} as a 2×2 block matrix with blocks $X = A_n$, $Y = V1_n$, $Z = V2_n$, and $\omega = b_0$. By the Frobenius identity for 2×2 block matrices, $\left\|\left(\begin{smallmatrix} X & Y \\ Z & \omega \end{smallmatrix}\right)\right\|_F^2 = \|X\|_F^2 + \|Y\|_2^2 + \|Z\|_2^2 + |\omega|^2$. Substituting the blocks gives (1). □

Lemma 1 (Frobenius norm for a 2×2 block). Let $M = \left(\begin{smallmatrix} X & Y \\ Z & \omega \end{smallmatrix}\right)$ with $X \in \mathbb{C}^{n\times n}$, $Y \in \mathbb{C}^{n\times 1}$, $Z \in \mathbb{C}^{1\times n}$ and $\omega \in \mathbb{C}$. Then

$$\|M\|_F^2 \;=\; \|X\|_F^2 + \|Y\|_2^2 + \|Z\|_2^2 + |\omega|^2.$$

Lemma 2 (Block inverse via Schur complement). If A is invertible, the Schur complement of A in $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ is shown below, where $S_D = D - CA^{-1}B$, and

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS_D^{-1}CA^{-1} & -A^{-1}BS_D^{-1} \\ -S_D^{-1}CA^{-1} & S_D^{-1} \end{pmatrix}.$$

(Proof widely available in the literature. The dual formula holds if D is invertible.)

Theorem 4 (Frobenius norm of A_{n+1}^{-1} in scalar-m_0 form).

$$A_{n+1} \;=\; \begin{pmatrix} A_n & V1_n \\ V2_n & b_0 \end{pmatrix}, \qquad A_n \in \mathbb{C}^{n\times n} \text{ nonsingular},\; V1_n \in \mathbb{C}^{n\times 1},\; V2_n \in \mathbb{C}^{1\times n},\; b_0 \in \mathbb{C}.$$

Define

$$k_0 := V2_n A_n^{-1} V1_n, \qquad m_0 := \frac{1}{b_0 - k_0}, \qquad V := V1_n V2_n, \qquad L_n := I_n + m_0\, V\, A_n^{-1}.$$

(So $b_0 - k_0 \neq 0$ is assumed, i.e. A_{n+1} is invertible.) Then

$$\boxed{\left\|A_{n+1}^{-1}\right\|_F \;=\; \sqrt{\left\|A_n^{-1}L_n\right\|_F^2 \;+\; |m_0|^2\Big(\left\|A_n^{-1}V1_n\right\|_2^2 + \left\|V2_nA_n^{-1}\right\|_2^2 + 1\Big)}\;.} \tag{2}$$

Proof. Apply the block inverse formula (Lemma 2) to $A_{n+1} = \left(\begin{smallmatrix} A_n & V1_n \\ V2_n & b_0 \end{smallmatrix}\right)$ with $A = A_n$, $B = V1_n$, $C = V2_n$, $D = b_0$. The Schur complement is

$$S = b_0 - V2_nA_n^{-1}V1_n = b_0 - k_0, \qquad \text{so} \qquad M := S^{-1} = m_0 \text{ (a scalar)}.$$

Therefore

$$A_{n+1}^{-1} = \begin{pmatrix} A_n^{-1} + A_n^{-1}V1_n m_0 V2_n A_n^{-1} & -A_n^{-1}V1_n m_0 \\ -m_0 V2_n A_n^{-1} & m_0 \end{pmatrix}.$$

Factor the top-left block as

$$A_n^{-1} + A_n^{-1} V1_n m_0 V2_n A_n^{-1} = A_n^{-1}\big(I_n + m_0\, V\, A_n^{-1}\big) = A_n^{-1} L_n.$$

Now use the Frobenius norm identity for a 2×2 block (Lemma 1) with

$$X = A_n^{-1} L_n, \quad Y = -A_n^{-1} V1_n m_0, \quad Z = -m_0 V2_n A_n^{-1}, \quad \omega = m_0.$$

We obtain

$$\big\|A_{n+1}^{-1}\big\|_F^2 = \big\|A_n^{-1} L_n\big\|_F^2 + \big\|A_n^{-1} V1_n m_0\big\|_2^2 + \big\|m_0 V2_n A_n^{-1}\big\|_2^2 + |m_0|^2.$$

Since m_0 is scalar, $\|A_n^{-1} V1_n m_0\|_2^2 = |m_0|^2 \|A_n^{-1} V1_n\|_2^2$ and similarly for the $V2_n A_n^{-1}$ term, giving (2) after taking the square root. This completes the proof of Theorem 1. □

2 Numeric Examples

The example below illustrate for a near-singular 4x4 matrix shows how the 4×4 breakdown of C_{ij} pinpoints which interactions dominate the condition number.

Near-singular example (4×4) matrices, calculations and heatmap

To illustrate the decomposition, consider the near-singular matrix

$$A_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1.5 \end{bmatrix}.$$

Its Frobenius condition number is already very large,

$$\kappa_F(A_4) \;\approx\; 2.43 \times 10^3.$$

Partitioning gives

$$A_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \quad b_0 = 1.5.$$

From these blocks,

$$k_0 = V_2 A_3^{-1} V_1 \;\approx\; 1501.0, \quad m_0 = \frac{1}{b_0 - k_0} \;\approx\; -6.67 \times 10^{-4}.$$

The four-term Frobenius norms are

$$S_1 = \|A_3\|_F^2 + \|V_1\|_2^2 + \|V_2\|_2^2 + |b_0|^2 \;\approx\; 13.26,$$

$$S_2 = \|A_3^{-1} L_3\|_F^2 + |m_0|^2\big(\|A_3^{-1} V_1\|_2^2 + \|V_2 A_3^{-1}\|_2^2 + 1\big) \;\approx\; 4.44 \times 10^5.$$

Their product yields

$$\kappa_F(A_4) = \sqrt{S_1}\,\sqrt{S_2} \;\approx\; 2427.17,$$

in agreement with the direct computation.

Table 2: Heatmap of C_{ij} contributions (Near-singular example). Darker = larger.

Row \ Column	$\mathbf{A_n^{-1}L_n}$	$\mathbf{m_0A_n^{-1}V_{1n}}$	$\mathbf{m_0V_{2n}A_n^{-1}}$	$\mathbf{m_0}$
$\mathbf{A_n}$	4,002,378.7514	14.0307	14.0307	0.0000
$\mathbf{V_{1n}}$	444,412.2284	1.5579	1.5579	0.0000
$\mathbf{V_{2n}}$	444,412.2284	1.5579	1.5579	0.0000
$\mathbf{\lvert b_0\rvert}$	999,927.5140	3.5053	3.5053	0.0000

Above we see the table with sixteen contributions C_{ij}. The dark region in the row of A_n against the column of $A_n^{-1}L_n$ dominates the condition number, with substantial secondary weight from $|b_0|$. This pinpoints where instability concentrates in the decomposition.

Next we apply Theorem 1 to the special cases of a 2x2 and 3x3 matrices. For matrices 3x3 and over we have no simplification for the general matrix and Theorem 1 is the best choice

Corollary 4 (2×2 closed form)**.** *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(A) \neq 0.$$

Then the Frobenius condition number satisfies

$$\kappa_F(A) \;=\; \frac{\|A\|_F^2}{|\det(A)|} \;=\; \frac{|a|^2+|b|^2+|c|^2+|d|^2}{|ad-bc|}.$$

Proof. This follows from Theorem 1 with $n = 1$. □

Corollary 5 (3×3 specialization)**.** *Let*

$$A_3 = \begin{pmatrix} A_2 & V_1 \\ V_2 & b_0 \end{pmatrix}, \qquad A_2 \in \mathbb{K}^{2\times 2} \text{ invertible}, \; V_1 \in \mathbb{K}^{2\times 1}, \; V_2 \in \mathbb{K}^{1\times 2}, \; b_0 \in \mathbb{K}.$$

Define

$$k_0 := V_2 A_2^{-1} V_1, \qquad m_0 := \frac{1}{b_0 - k_0}, \qquad L_n := I_2 + m_0\,(V_1 V_2) A_2^{-1}.$$

$$\kappa_F(A_3) = \sqrt{\|A_2\|_F^2 + \|V_1\|_2^2 + \|V_2\|_2^2 + |b_0|^2}\;\sqrt{\|A_2^{-1}L_2\|_F^2 + |m_0|^2\big(\|A_2^{-1}V_1\|_2^2 + \|V_2A_2^{-1}\|_2^2 + 1\big)}\,.$$

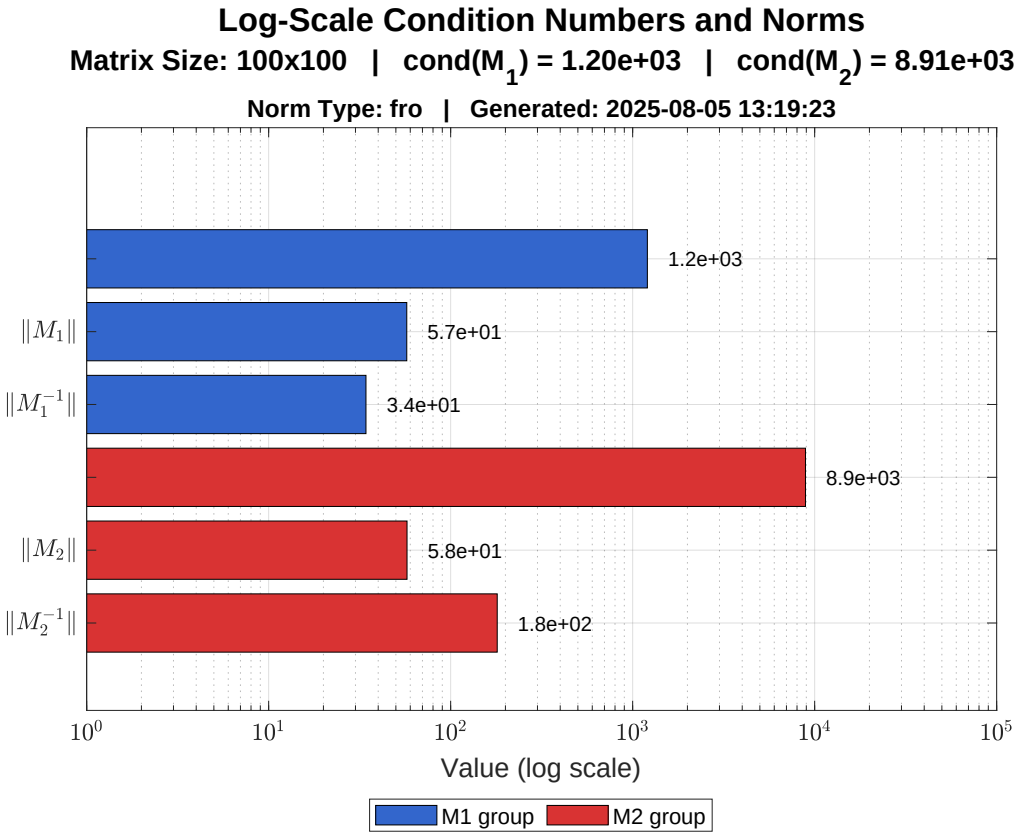


Figure 5: In this instance of two random matrices we see the breakdowns of the norms and the Frobenius condition numbers of the two groups, condition numbers are unlabeled in the chart.

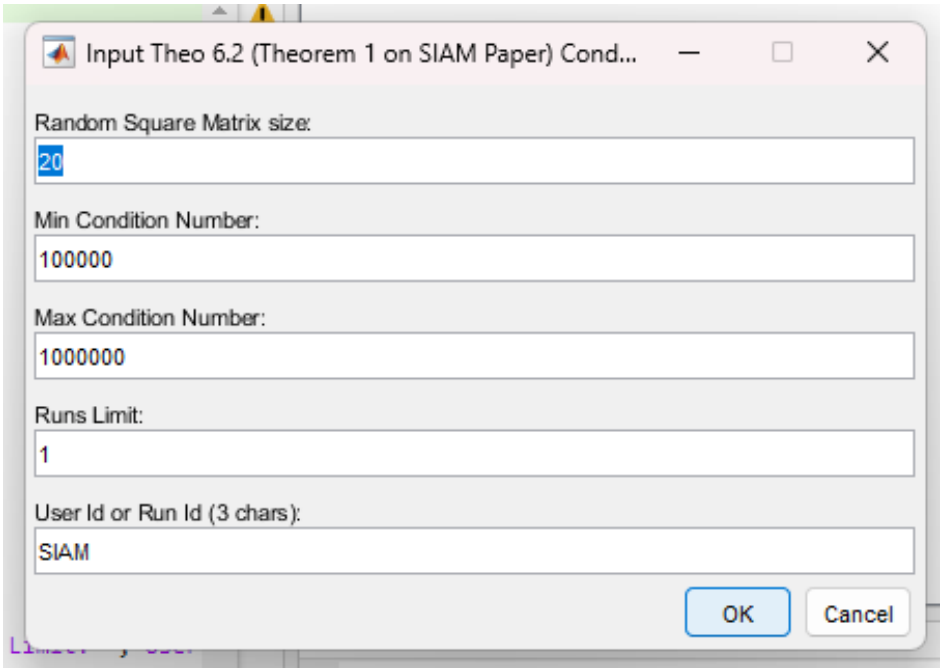


Figure 6: Sample of Matlab Prompt for our Theorem 1 decomposition code to specify 1-Random Matrix dimension 2- Min and Max Condition Number, How many runs to do and an user ID. Most of these are included in the plots title and saved in a directory based on run date etc.

Numerical Verification and Visualizations

To complement the theoretical results, we illustrate their behavior on representative matrices. The plots confirm both the accuracy of the closed form and the localization of sensitivity within the block L_n.

...

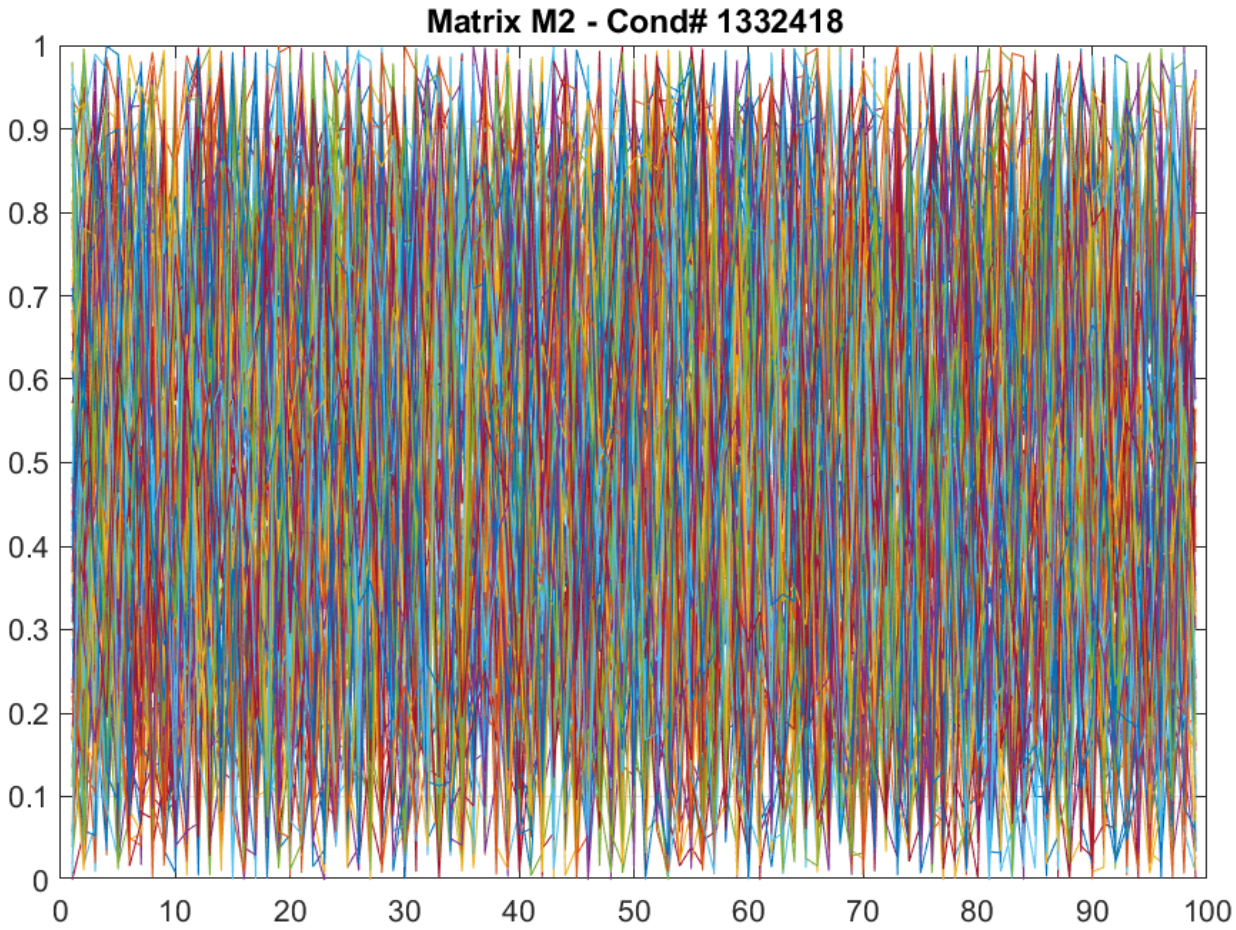


Figure 7: The M2 Matrix - For a random matrix with large cond. Number: 1,332,418

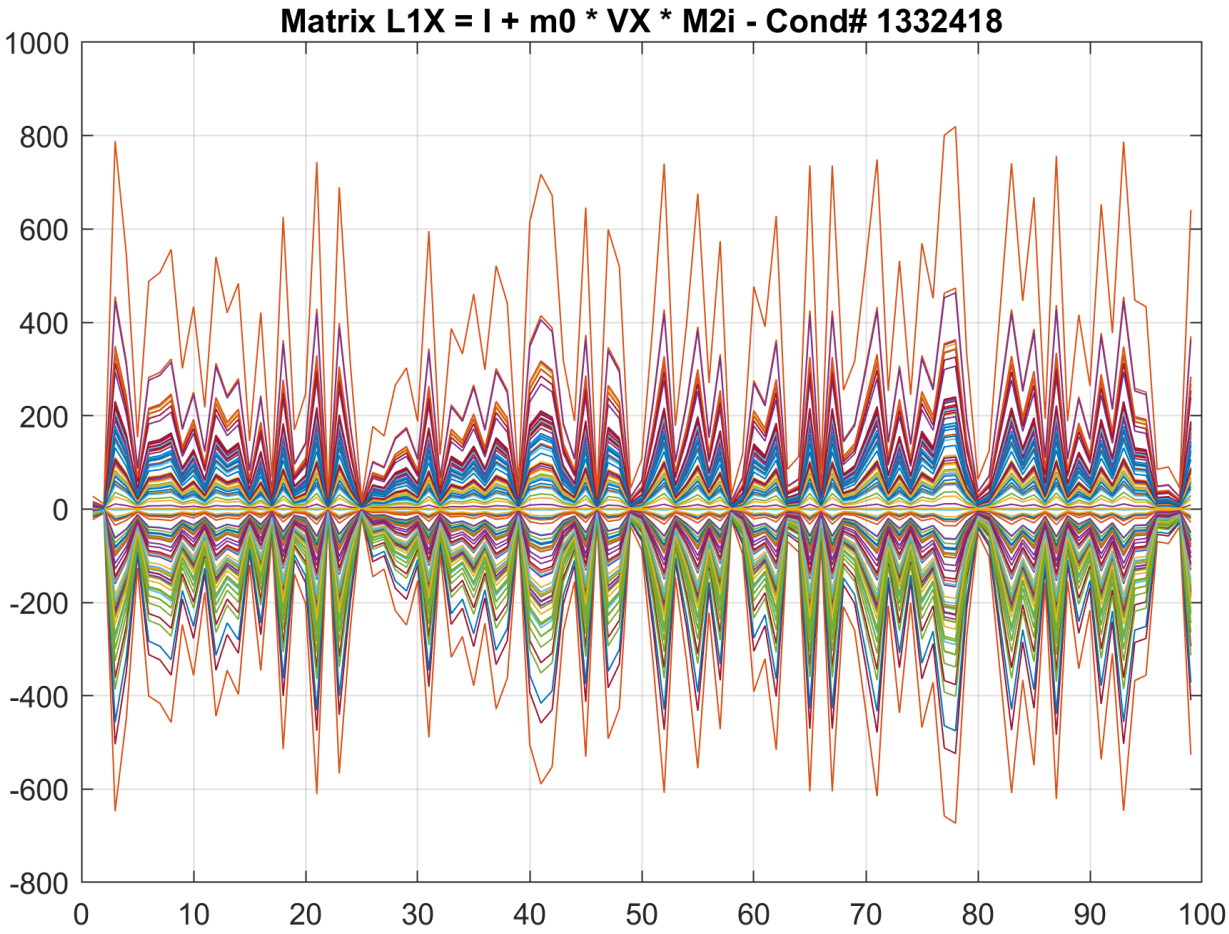


Figure 9: The Ln Matrix - For a random matrix with large cond. Number: 1,332,418

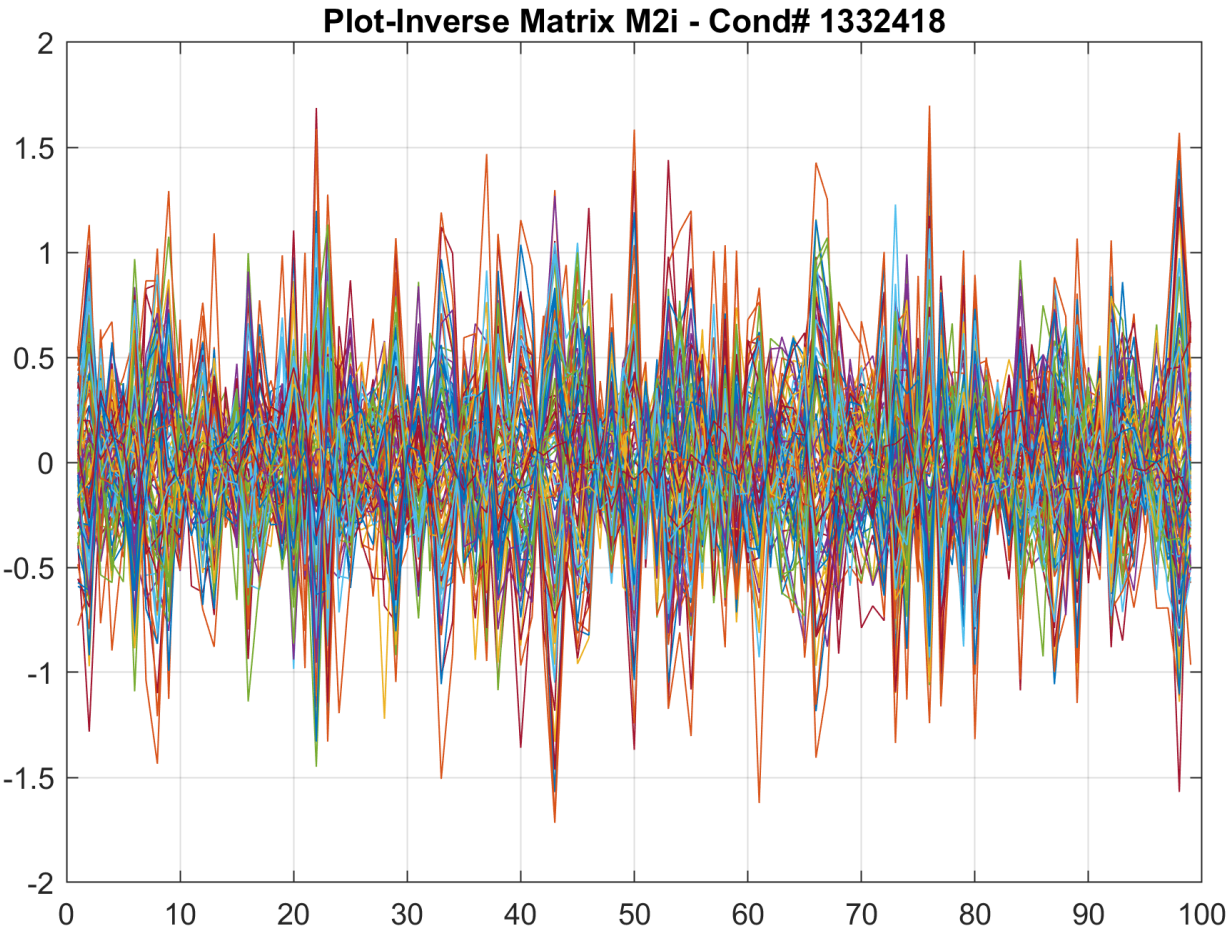


Figure 8: The M2 INVERSE Matrix - For a random matrix with large cond. Number: 1,332,418

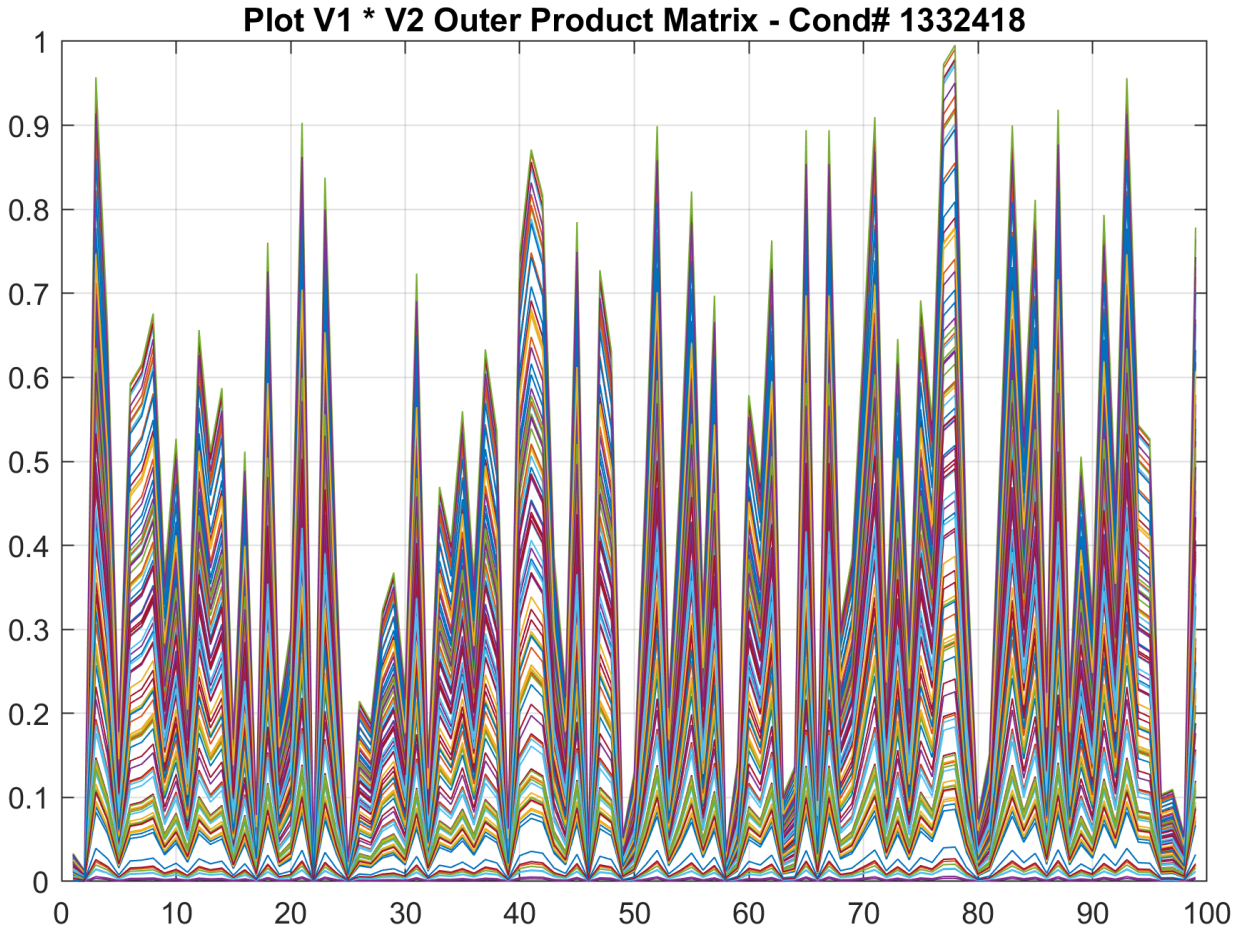


Figure 10: The Prod of Edge Vectors V1xV2 - For a random matrix with large cond. Number: 1,332,418

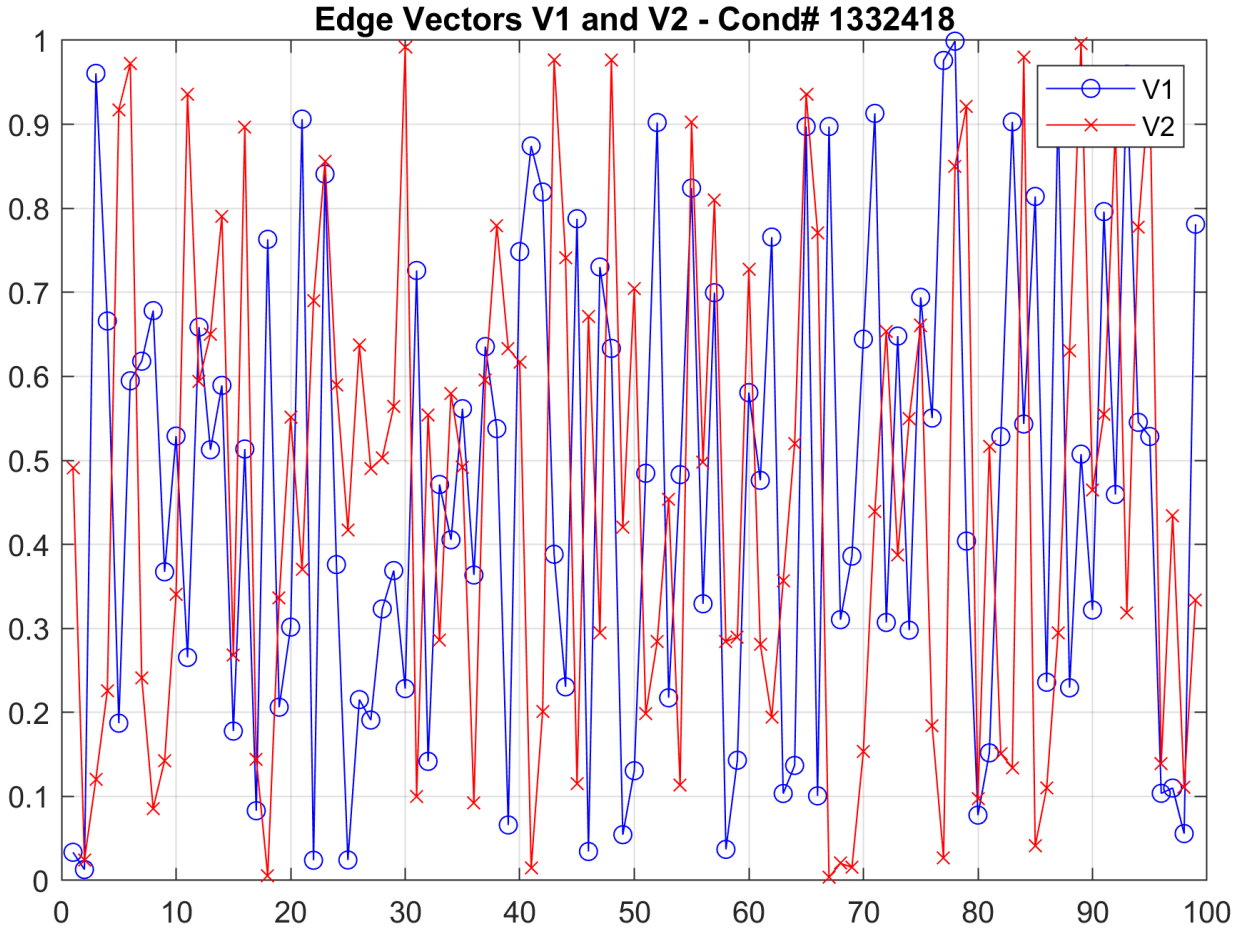


Figure 11: Edge Vectors V1 and V2 - For a random matrix with large cond. Number: 1,332,418

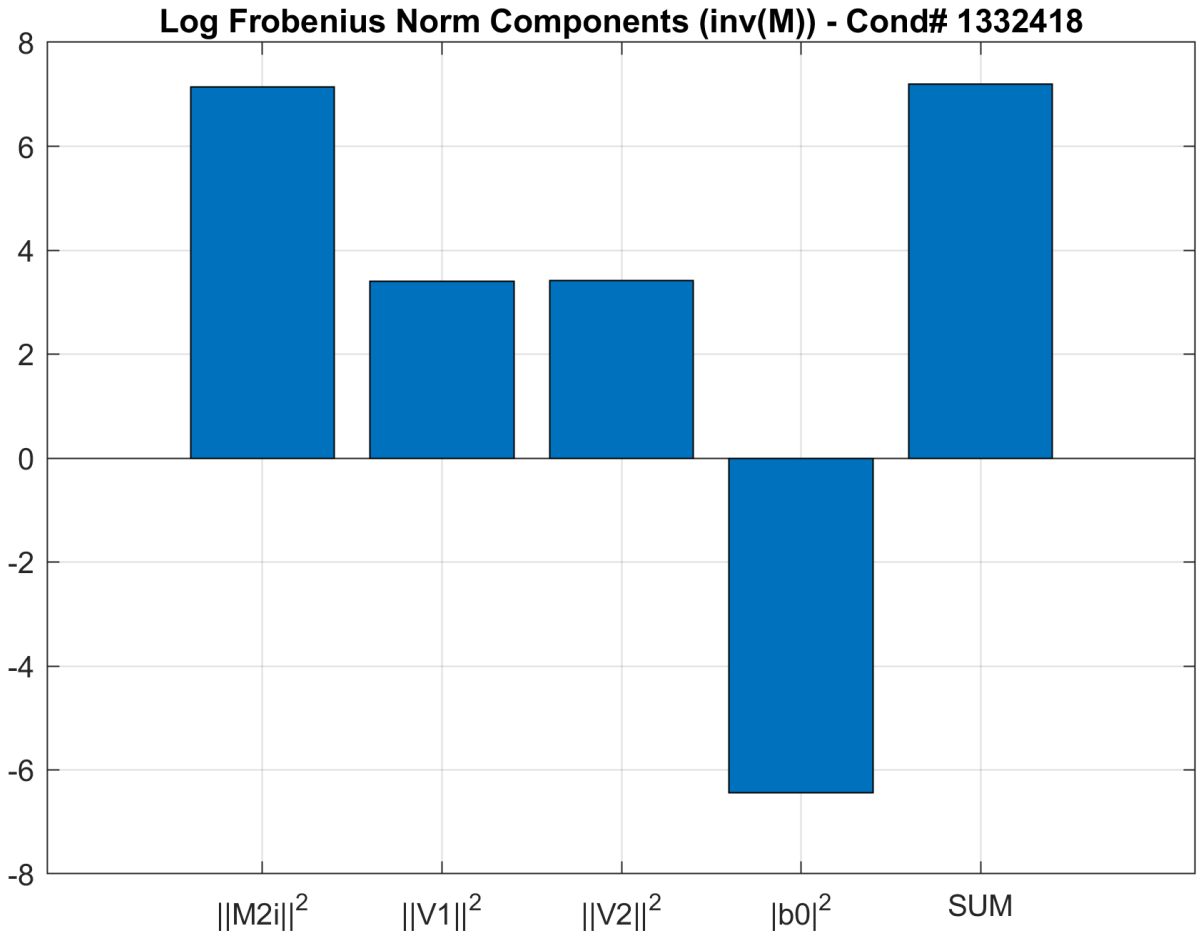


Figure 12: The Log of Frobenius Norms - For the Inv(M) with large cond. Number: 1,332,418

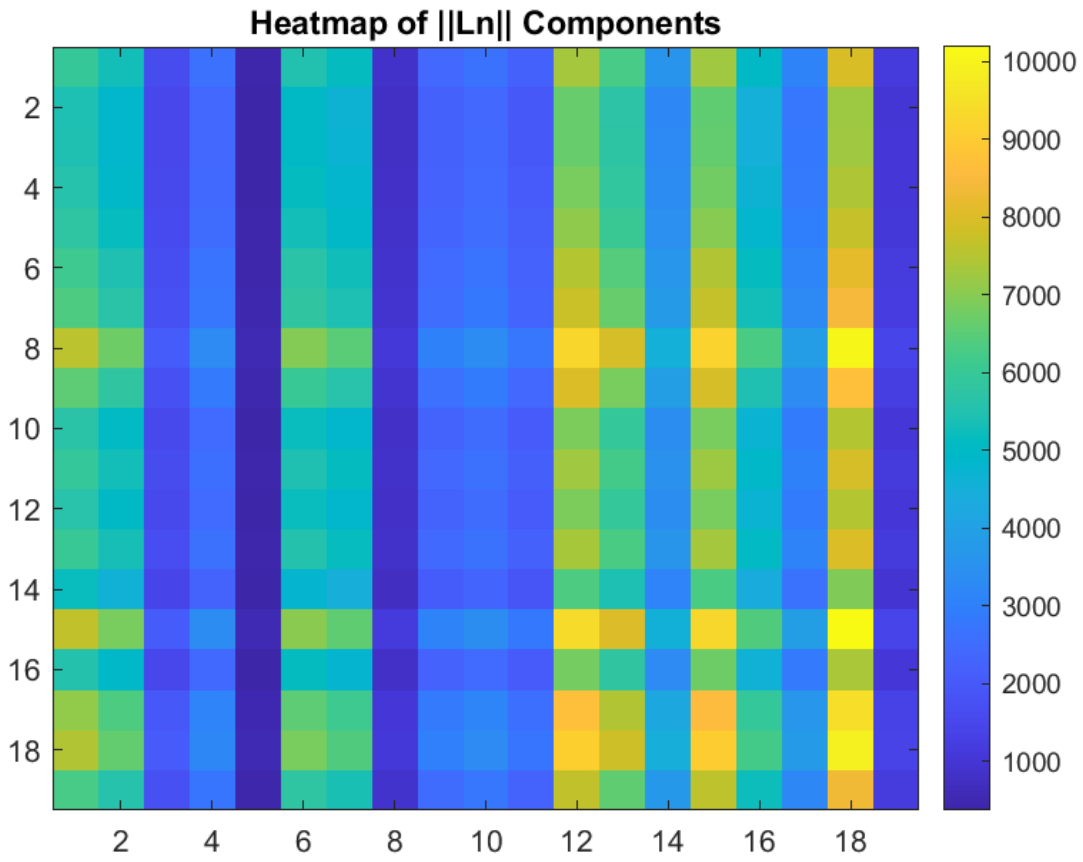


Figure 13: The heatmap plot of Ln - possibly the most complex matrix block: I+ m0V1V2xInv(An)

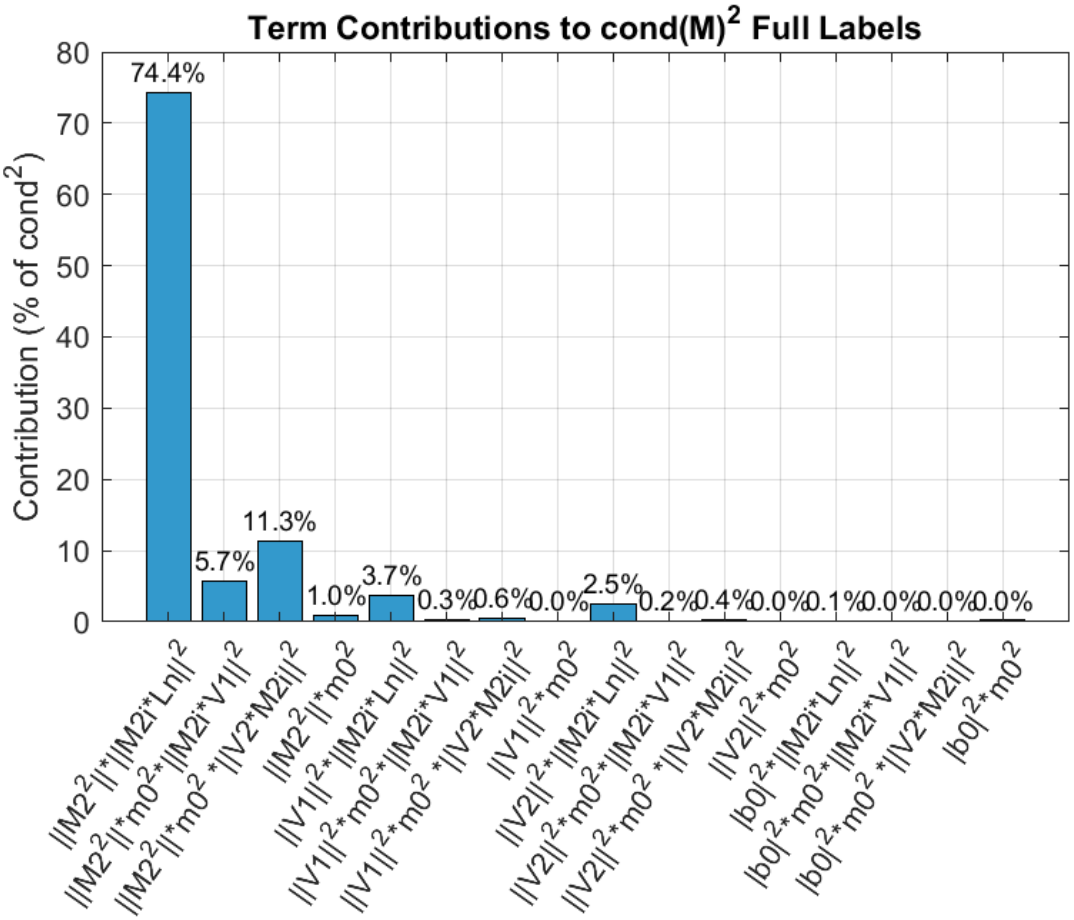


Figure 14: When we multiply the 4x4 Terms in the Frobenius Condition Number Squared, we get these 16 terms. Now we can address the question: What makes a large condition number?
First, observe that again the Ln matrix is in the tallest bar! Suggested procedure: Just look at the top bars for a given matrix (recall that the condition number will depend on the matrix structure and contents and its inverse matrix too) and then analyze the norms below it. Note that all norms are the Frobenius norms. For scalars its the absolute value (Real or Complex) squared. The letter i stands for the inverse matrix. There are errors in some powers of 2 which seem the power of 2 should be outside the norm symbol.

```
normType = 'fro' % Frobenius norm only

%  d = matrix dimension, below we generate the matrices
M1 = rand(d);   % An
M2 = rand(d+1); % An+1

% next we load the submatrix M1 also into the 2nd larger matrix M2
M2(1:d,1:d) = M1; % that is M1 is contained in M2

% compute each matrix inverses for A
M1i = inv(M1);
M2i = inv(M2);


% the edge vecntors for M2
V1 = M2(1:d,d+1);
V2 = M2(d+1,1:d);

% the prod of V1 and V2 a nxn matrix
VX = V1 * V2;

% the all important constants: k0, b0 (last diagonal in An+1, and m0)
k0 = V2 * M1i * V1;
b0 = M2(d+1,d+1);
m0 = 1 / (b0 - k0);

% Identity nxn matrix for Ln
I = eye(d);
% The inverse of An times Ln matrix into L1
L1 = M1i * (I + m0 * VX * M1i);

% Interemediate matrices related to edge vectors and the inverse
F1 = M1i * V1;
F2 = V2 * M1i;

format shortG %MAKE TITLES SHORTER & CLEAR

% === Scalars & Chart Title Metadata ===
condM1 = cond(M1,normType); Frobenius norm
normM1 = norm(M1,normType); Frobenius norm

titleInfo = ['ID: ' num2str(runID) ...
             ' | UserID: ' num2str(userId) ...
             ' | DIM: ' num2str(d) ...
             ' | CN: ' num2str(condM1,'%.0f') ...
             ' | NORM: ' num2str(normM1,'%.0f')];
```

Figure 15: The core code in Matlab (matrices M-file language) for the computations of the Theorem 1 CANMXP decomposition. This code can be usually embedded in a loop of runs, often we generate a random square matrix and select those that exceed a user-defined minimum (fro) Condition Number for research, or are are with a min/max range. Note how we generate two random matrices M1 and M2 (the larger matrix), but then insert into M1 the matrix M2 so both share the same submatrix of dimension dxd. For applying this to a given non-random matrix we modify this code to load the matrix (in the workspace or from a file) and skip the 1st 3 steps.

Finite-precision effects and sources of error in matrix computations

Most production systems (CPUs, GPUs, clusters) implement IEEE 754 floating–point arithmetic. In binary64 ("double"), machine epsilon is $\varepsilon \approx 2^{-53} \approx 1.11 \times 10^{-16}$. Arithmetic is correctly rounded but *not associative or distributive in finite precision*, and mixed–precision paths (e.g., fp32/fp16/bfloat16 with accumulation in fp32) are common for speed. Consequently, small backward errors $\mathcal{O}(\varepsilon)$ can be amplified by the condition number, yielding forward errors on the order of $\kappa\,\varepsilon$.

Typical causes of error in matrix operations.

- **Ill-conditioning (problem level).** When $\kappa(A)$ is large, any backward error from rounding is magnified in the solution (*stable algorithm* $\Rightarrow$ forward error $\lesssim \kappa(A)\,\varepsilon$).
- **Cancellation and loss of significance.** Subtractions of nearly equal quantities (e.g., forming normal equations $A^\top A$) destroy significant digits and can square the condition number relative to A.
- **Pivot growth and factorization sensitivity.** In Gaussian elimination without adequate pivoting, entries can grow before division, increasing rounding error; analogous issues arise in QR/orthogonalization variants with loss of orthogonality (e.g., classical Gram–Schmidt).
- **Summation / reduction order.** Floating–point sums depend on evaluation order; parallel reductions and tree sums can change accumulated roundoff by orders of magnitude for long vectors.
- **Scaling and dynamic range.** Very large/small entries risk overflow/underflow; subnormal (flush–to–zero) behavior and denormal penalties can alter effective arithmetic.
- **Mixed precision and conversions.** Casting between precisions (fp16$\leftrightarrow$fp32, bfloat16, tensor–core FMAs) introduces quantization and rounding asymmetries.
- **Nonassociativity in blocked/parallel kernels.** Reordering due to tiling, blocking, and thread scheduling changes rounding paths (even with identical inputs).
- **Derivative propagation.** In optimization/ML, Jacobian/Hessian builds and backprop compose many linear ops; rounding errors accumulate and are amplified where the intermediate operators are ill–conditioned.

Applications and Relevance to High-Performance Computing, Machine Learning, Artificial Intelligence, and Mathematical Finance

Role of the decomposition. Theorem 1 expresses the Frobenius–norm condition number of a block matrix as a product of two four-term factors, producing a 4×4 table of contributions. This makes visible which subblocks dominate ill-conditioning without prescribing a specific algorithmic fix. In practice, it shows *where* sensitivity originates before one chooses preconditioners, scalings, or block reorderings.

High-Performance Computing (HPC). Large-scale linear solvers (e.g., Krylov and multigrid methods) are governed by matrix conditioning. The 4×4 breakdown often reveals concentration in terms such as $A_n^{-1} L_n$, guiding domain decomposition interfaces, block reorderings, and where a preconditioner should act most strongly. Because the ingredients are Frobenius norms and solves with A_n, the diagnostic is less expensive than full spectral analyses and scales naturally to block-structured systems.

Machine Learning (ML) and Artificial Intelligence (AI). Training pipelines (least squares, kernel methods, and gradient propagation through linear layers) implicitly solve large linear systems. Ill-conditioning amplifies data noise and destabilizes updates. The blockwise view distinguishes whether instability stems from the core submatrix A_n, the edge couplings V_1, V_2, or the pivot b_0, supporting practical decisions such as data scaling, feature selection, or constraint auditing.

Numerical Optimization. Convergence of Newton and quasi-Newton methods depends on the conditioning of Hessian or Jacobian normal equations. When such matrices are assembled in blocks, the 4×4 map pinpoints "trouble" interfaces and mixed terms that slow convergence or cause divergence.

Mathematical Finance (q-fin.MF). Covariance and correlation matrices used in portfolio construction and risk management can be extremely ill-conditioned. The decomposition clarifies whether instability is driven by market-wide structure (the main block), sector couplings (edge interactions), or single-asset pivots. This is directly relevant to high-dimensional problems such as S&P 500 covariance estimation, where preconditioning or factor-model choices strongly affect stability.

Scope. The present note provides diagnosis, not remedies. Algorithmic consequences (targeted preconditioning or automated block selection) and large-scale benchmarks are deferred to a separate implementation-focused paper.

Acknowledgments

First, my great appreciation for the great math teachers I had while in the University of Puerto Rico (UHS, UPR and UPRM) and the University of California - San Diego (UCSD).

Code availability

A repository with the code index and usage notes supporting this work is available at: `https://github.com/codetigers314159/AMMA-MatrixConditionNumberTheorems`. This paper refers to the index rather than specific filenames; the version scheme is described in the README.